



Sanskar Nanda

✉ sanskarnanda04@gmail.com | 📞 +91 9776631525 | 🌐 SanskarNanda

Education

IISER Berhampur

Integrated BS–MS Dual Degree in Physics

- CPI: 8.3 / 10

2021 – 2026

Berhampur, India

ODM Public School

Class 12th, Central Board of Secondary Education (CBSE)

- Percentage: 83.83%

2018 – 2020

Bhubaneswar, India

St. Joseph's High School

Class 10th, Indian Certificate of Secondary Education (ICSE)

- Percentage: 92.66%

2016 – 2018

Bhubaneswar, India

Master Thesis Project

Collider-Independent Asymmetries in Top-Quark Pair Production

Supervisor: Prof. Gagan B. Mohanty, TIFR Mumbai

Co-Supervisor: Dr. Md Nasim, IISER Berhampur

- **Objective:** Unlike the collider-dependent observables A_{FB} (Tevatron) and A_C (LHC), this work focuses on measuring the intrinsic parton-level asymmetries A_u and A_d in $t\bar{t}$ production. These observables are independent of the collider type and are directly sensitive to potential BSM scenarios, as discussed in the literature.
- **Validation and Background estimation:** Validated the Data–MC agreement after applying relevant MC corrections and estimated the QCD multijet background using a data-driven ABCD method.
- **ML implementation:** Since the $t\bar{t}$ asymmetry arises from the $q\bar{q} \rightarrow t\bar{t}$ channel at NLO, a BDT is trained to distinguish it from the dominant $gg \rightarrow t\bar{t}$ background.
- **Reconstruction and Unfolding:** Reconstruction of the $t\bar{t}$ system using a constrained χ^2 minimization technique and unfolding with TUnfold to extract the collider-independent asymmetry observables.
- **Tools used:** Coffea, Python, C++, Dask, ROOT, PyROOT
- **Thesis:** 📄 Master Thesis
- **Repository:** 🌐 TOP-ANALYSIS

Internships

Three-Particle Azimuthal Correlations

May 2024 – August 2024

Supervisor: Dr. Prabhat Ranjan Pujahari, IIT Madras

- Analyzed **three-particle azimuthal correlations** in pp and Pb-Pb collisions.
- Implemented **mixed-event techniques** to isolate genuine multi-particle correlations.
- Performed a comparative study between **pp and Pb–Pb collisions**, analyzing near-side ($\Delta\phi = 0$) and away-side ($\Delta\phi = \pi$) peaks for different charge combinations of the trigger and associated particles.
- **Tools Used:** ROOT, CMSSW, C++, Python

Shack-Hartmann Wavefront Sensor Development

Dec 2022 – Feb 2023

Supervisor: Dr. Vyas Akondi, IISER Berhampur

- Surveyed Shack-Hartmann lenslet and pixel requirements for a prototype module
- Reviewed lenslet geometries and pixel specs for prototype wavefront sensor.

Technical Skills

- **HEP Frameworks:** ROOT, CMSSW, Coffea, Pepper
- **Programming and Scripting:** C++, Python, Java, LaTeX, Bash
- **Statistical Tools:** Combine, RooFit
- **Operating Systems:** Linux, Windows
- **Development Tools:** Git/GitHub
- **Distributed Computing:** Dask
- **Machine Learning:** BDT, ANN, GNN, CNN, transformers

Workshops

ML4HEP-V4 School

Pre and Main School

TIFR Mumbai

29th June–11th July, 2026

- Selected among 35 participants from around 350 applicants for the ML4HEP Main School.
- Completed lectures and hands-on sessions on statistical methods, CNNs, GNNs, GCNs, and Transformers.
- As part of the final team project on electron energy regression in an HGCAL prototype, I implemented baseline (non-ML) and Deep Neural Network (DNN) models, while my teammates developed BDT-, GCN-, and Transformer-based models. The performance of the different approaches was evaluated through energy resolution studies, with the Transformer-based model achieving the best overall performance.
-  **ML4HEP-Certificate**

CMS Data Analysis School (CMSDAS 2025)

Long and Short Exercises

IIT Hyderabad

23rd–27th June, 2025

- Collaborated in a team to measure the $t\bar{t}$ production **cross section** in the $e\mu$ channel. Developed analysis pipelines leveraging pepper, coffea, uproot, and awkward for event selection, implementing MC corrections and data–MC validation.
- Learned to interpret impact plots generated using CMS Combine, to study how the systematics influence the measurement.
- Completed short exercises on *EGamma*, *Jets+MET*, *BTV*, and *Statistics & Combine*.
- Our team was awarded **Best Group Presentation** for clarity of analysis and presentation.
-  **CMSDAS-LONG-EXERCISE-PRESENTATION**

CNI Summer School 2023

Dynamic Resource Allocation in Communication Networks

IISc Bangalore

10th–14th July, 2023

- Gained experience with mathematical and computational techniques for dynamic resource allocation, including Markov decision processes, LP-based heuristics, Whittle index policies, and Deep Reinforcement Learning.
- Explored applications to network problems such as adaptive routing, TCP control, age-of-information optimization, and channel selection through guided Python lab sessions implementing the studied algorithms.
-  **CNI-Summer-School-Certificate**

DST–Rajasthan IASC

International Astronomical Search Collaboration

IISER Berhampur

18th Nov–13th Dec, 2022

- Analyzed astronomical images from the **Pan-STARRS** using **Astrometrica** software to observe Near-Earth Objects (NEOs) and main-belt asteroids.
-  **DST–Rajasthan IASC Certificate**

Science Outreach

Volunteer, National Science Day (NSD), TIFR Mumbai

February 2026

- Guided college students and members of the general public through active research laboratories at TIFR Mumbai as part of National Science Day, facilitating interactions with researchers and providing exposure to cutting-edge experiments and ongoing scientific developments across diverse fields.

Jigyansa – Outreach Council, IISER Berhampur

August 2022 – May 2024

- Member of IISER Berhampur's student-led science outreach council, contributing as a content writer for the outreach team by preparing scientific articles for the general public and explaining complex concepts in a clear and accessible manner.

Language Proficiency

- English
- Hindi
- Odia (Native)

Achievements

National Level Examinations

- Qualified **IAT (IISER Aptitude Test)**
- Qualified **JEE (Joint Entrance Examination)** with a score of **87 percentile**
- Qualified **OUAT (Odisha University of Agriculture and Technology)** with a score of **95 percentile**

References

- **Prof. Gagan B. Mohanty – Professor, Department of High Energy Physics**
Tata Institute of Fundamental Research (TIFR), Mumbai
Email: gmohanty@tifr.res.in
- **Dr. Md. Nasim – Associate Professor, Department of Physical Sciences**
Indian Institute of Science Education and Research, Berhampur
Email: nasim@iiserbpr.ac.in
- **Dr. Prabhakar Palni – Assistant Professor, School of Physical Sciences**
Indian Institute of Technology, Mandi
Email: prabhakar@iitmandi.ac.in